

MD MASHFIQ RIZVEE

Lawrence, Kansas, USA

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RESEARCH INTEREST

My research interests include artificial intelligence, computer vision, biometric authentication, data privacy and applied data science. My current focus is on scalable and trustworthy biometric systems for human-centered security, privacy-preserving identity verification, Bloom filter-based authentication and secure data management.

ACADEMIC CREDENTIALS

University of Kansas, USA *May 2027 (Expected)*
Pursuing Doctor of Philosophy, Computer Science

University of Kansas, USA *Aug. 2022 – May 2025*
Master of Science, Computer Science
Cumulative GPA: 4.00/4.00

North South University, Bangladesh *Jan. 2018 – Dec. 2021*
Bachelor of Science, Computer Science and Engineering
Cumulative GPA: 3.91/4.00

WORK EXPERIENCE

Jumio Corporation, USA *Apr. 2026 – Present*
Role: Data Engineering Intern

- Developed a production ready state-of-the-art DeepFake detection framework leveraging Fast Fourier Transform (FFT)-based frequency-domain artifacts to identify subtle inconsistencies introduced during synthetic face generation and manipulation, with a focus on improving detection accuracy and generalization regardless of generation methods.
- Engineered scalable data acquisition, synthetic user simulation, and dataset generation pipelines for biometric verification systems. Integrated nested event parsing, procedural scenario coverage, behavioral signal extraction, and data-quality estimation from real-world authentication event data for liveness.
- Extracted and analyzed authentication telemetry signals, including SDK versions, device characteristics, motion analysis, liveness scores, and photinus features, for model benchmarking and experimentation.
- Developed large-scale data querying pipelines using PySpark, Spark SQL, and EMR/S3-based infrastructure to retrieve and analyze synthetic biometric authentication events for liveness fraud analysis.
- Collaborated on workflows supporting reward-informed model evaluation, verification-guided experimentation, and production training/evaluation pipelines, with applications in human authentication, anti-spoofing, multilingual/generalized robustness, and AI safety-oriented model testing.

University of Kansas, USA *Aug. 2022 – Present*
Role: Graduate Research Assistant

- Developed efficient biometric database system designed to track humans using facial biometrics. Ensured enhanced security measures as well as established a comprehensive security framework for the database system, incorporating encryption, authentication, and access control mechanisms.
- Developed a secure database system designed to track a high volume of semiconductors within an electronic supply chain using PUFs. Evaluated results using SPC and DOE methods for monitoring variations over time. Collected PUF data from Ring-Oscillator-based RO-PUF with a MicroBlaze microprocessor deployed in the Basys-3 FPGA.
- Evaluated the security of biometric database systems by analyzing vulnerabilities against enrollment attacks, query attacks, denial-of-service (DoS) attacks, presentation attacks, and pre-image attacks. Documented threat models and mitigation strategies to strengthen authentication reliability across biometric and hardware-security deployments.
- Developed efficient CNN architectures and image classifiers that reduce the carbon footprint and energy consumption of AI systems. This work resulted in a patent.

- Collaborated with the Civil Engineering Department and developed a real-time road paving machine vision tool using OpenCV and FastSAM, enabling large CAT paver machines to detect paving boundaries and monitor asphalt distribution for construction safety. A pilot project was launched with the help of the City of Lawrence.

University of Kansas, USA

Spring 2025, Spring 2026

Role: Graduate Teaching Assistant

- Instructed students through hands-on PCB soldering tasks in lab sessions, covering basic hardware assembly and circuit troubleshooting for digital circuit design principles.
- Guided students in using Xilinx Vivado for software-based labs involving logic building, simulation, synthesis, and implementation of digital circuits using VHDL and Verilog.

Texas Tech University, USA

Jan. 2022 – Aug. 2022

Role: Graduate Teaching Assistant

- Assisted instructors by evaluating exam scripts and assignments and providing additional student support for Computer Systems Architecture and Organization and Cryptography courses.
- Prepared inception documents, project-learning documents, and other project monitoring reports.

North South University, Bangladesh

Jan. 2021 – Dec. 2021

Role: Undergraduate Teaching Assistant

- Assisted faculty with course support, grading, student guidance, and academic documentation for computer science coursework.

PROJECTS

FarmWise: Your Farming Companion

Hack Midwest, 1st Place

- Built accessible user interfaces through Flask and Streamlit, incorporating AWS Bedrock and LLM-based contributions. Achieved Best Project Award, supporting USDA initiatives related to Artificial Intelligence Strategy.

Automated Detection of Counterfeit Semiconductors at Scale

HOST 2022, Winner

- Developed an OpenCV- and machine learning-based inspection pipeline for scalable counterfeit semiconductor detection, focusing on automated analysis of integrated circuit markings and visual features for hardware security validation.

Quantifying Imbalanced Classification Methods for Leukemia Detection

- Implemented advanced sampling methods, including GAN-based augmentation, to refine dataset analysis, leading to critical insights that informed effective data visualization strategies and clarified blood cell classification nuances.
- Conducted controlled model comparison experiments equivalent to A/B testing using TensorFlow on imbalanced leukemia classification datasets. Evaluated sampling strategies such as GAN-based augmentation and achieved approximately 8% improvement in predictive accuracy, supporting data-driven model selection.

ZkPoH: Zero Knowledge Proof of Humanness

Hack Midwest Block-A-Thon, 1st Place

- Built a privacy-preserving biometric identity analytics pipeline using Polygon zkEVM, processing biometric embedding datasets and applying Hierarchical Bloom Filters as a smart contract for scalable storage and identity matching.
- Developed Rust backend services with ark cryptographic libraries and Solidity smart contracts deployed via Hardhat, enabling time-bound identity credentials stored on blockchain. Designed controlled performance experiments comparing authentication latency and storage efficiency across different Bloom filter configurations, achieving faster query response times and reduced memory overhead.

PUBLICATIONS

1. **Md Mashfiq Rizvee**, Domenic Forte, and Sumaiya Shomaji. “Open-Set Biometric Watchlist with Learning-Guided Bloom Filters,” Under review.
2. **Md Mashfiq Rizvee**, Cameron Denton, Domenic Forte, and Sumaiya Shomaji. “Bio-BloomChain: A Probabilistic Smart Contract for Secure and Scalable Decentralized Biometric Authentication Devices.” Under review.
3. **Md Mashfiq Rizvee**, Fairuz Shadmani Shishir, Tanvir Hossain, Tamzidul Hoque, Domenic Forte, and Sumaiya Shomaji. “A Persistent Hierarchical Bloom Filter-based Framework for Scalable Authentication and Tracking of ICs.” ACM Journal on Emerging Technologies in Computing Systems, 2025.

4. **Md Mashfiq Rizvee**, Pallabi Ghosh, Domenic Forte, and Sumaiya Shomaji. “Dynamic Hierarchical Bloom Filters for Scalable Biometric Authentication Systems.” International Joint Conference on Biometrics, 2025.
5. **Md. Mashfiq Rizvee**, Md. Amiruzzaman, and Md. Rajibul Islam. “Data Mining and Visualization to Understand Accident-prone Areas.” In International Joint Conference on Advances in Computational Intelligence. Springer, 2021.
6. Depto, Deponker Sarker, **Md Mashfiq Rizvee**, Aimon Rahman, Hasib Zunair, M. Sohel Rahman, and M. R. C. Mahdy. “Quantifying imbalanced classification methods for leukemia detection.” Computers in Biology and Medicine, 152, 106372, 2023.
7. **Md Mashfiq Rizvee**, Md Hafizur Rahman, Sumaiya Shomaji, and Prabuddha Chakraborty. “Understanding the Innovations Required for a Green and Secure Artificial Intelligence Paradigm.” 2023 IEEE 16th Dallas Circuits and Systems Conference, DCAS.
8. Md Hafizur Rahman, Zafaryab Haider, **Md Mashfiq Rizvee**, Sumaiya Shomaji, and Prabuddha Chakraborty. “Intelligent Layer Sharing ILASH: A Predictive Neural Architecture Search Framework for Multi-Task Applications.” IEEE Access, 2025.
9. Rokunuzjahan Rudro, Saumik Sakib Bin Masud, **Md Mashfiq Rizvee**, Shuyao Wang, Vishal C. Kumetha, Alexandra Kondyli, Christopher H. Ramey, Evangelia G. Chrysikou, and Sumaiya Shomaji. “Using Machine Learning to Classify Driver Behavior from Psychological Features: An Exploratory Study.” Transportation Research Board, TRB, 2025.

PATENTS

1. “Methods and systems for the design and use of intelligent layer shared neural networks and model architecture search algorithms,” WO2025049978A1, Date of Filing: 08.30.2024.
2. “BloomChain,” Application No.: 63/772,943, Date of Filing: 03.17.2025.
3. “Dynamic Hierarchical Bloom Filters,” Application No.: 63/772,943, Date of Filing: 03.17.2025.

PH.D. DISSERTATION

Title: Hierarchical Probabilistic Architectures for Scalable Biometric and Electronic Authentication in Secure Surveillance.

Committee: Dr. Sumaiya Shomaji, Dr. Tamzidul Hoque, Dr. Hongyang Sun, Dr. Alexandra Kondyli, Dr. David Johnson.

Summary: Secure and scalable authentication has become a primary requirement in modern digital ecosystems, where both human biometrics and electronic identities must be verified under noise, large population growth and resource constraints. Existing approaches often struggle to simultaneously provide storage efficiency, dynamic updates and strong authentication reliability. The proposed work advances a unified probabilistic framework based on Hierarchical Bloom Filter (HBF) architectures to address these limitations across biometric and hardware domains. The first contribution establishes the Dynamic Hierarchical Bloom Filter (DHBF) as a noise-tolerant and dynamically updatable authentication structure for large-scale biometrics. Unlike static Bloom-based systems that require reconstruction upon updates, DHBF supports enrollment, querying, insertion and deletion without structural rebuild. Experimental evaluation on 30,000 facial biometric templates demonstrates 100% enrollment and query accuracy, including robust acceptance of noisy biometric inputs while maintaining correct rejection of non-enrolled identities. These results validate that hierarchical probabilistic encoding can preserve both scalability and authentication reliability in practical deployments. Building on this foundation, Bio-BloomChain integrates DHBF into a blockchain-based smart contract framework to provide tamper-evident, privacy-preserving biometric lifecycle management. The system stores only hashed and non-invertible commitments on-chain while maintaining probabilistic verification logic within the contract layer. Large-scale evaluation again reports 100% enrollment, insertion, query and deletion accuracy across 30,000 templates, therefore, solving the existing problem of blockchains being able to authenticate noisy data. Moreover, the deployment analysis shows that execution on Polygon zkEVM reduces operational costs by several orders of magnitude compared to Ethereum, therefore, bringing enrollment and deletion costs below \$0.001 per operation which demonstrate the feasibility of scalable blockchain biometric authentication in practice. Finally, the hierarchical probabilistic paradigm is extended to electronic hardware authentication through the Persistent Hierarchical Bloom Filter (PHBF). Applied to electronic fingerprints derived from physical unclonable functions (PUFs), PHBF demonstrates robust authentication under environmental variations such as temperature-induced noise. Experimental results show zero-error operation at the selected decision threshold and substantial system-level improvements as well as over 10^5 faster query processing and significantly reduced storage requirements compared to large scale tracking.

PROPOSAL WRITING EXPERIENCE

CAREER: AuthenTrack: Secure and Scalable Frameworks for Large-Scale Authentication and Tracking

- Contributed substantially to the primary drafting and full manuscript revision as a student contributor. The proposal was closely aligned with my thesis research, incorporating related technical results and research outcomes. The proposal was successful and secured \$590,000 in funding for my PI, Dr. Sumaiya Shomaji.

Towards Building a Research Program Focused on Tracking and Surveillance of Hardware and Human

- Contributed to a proposal connected to my thesis research on tracking and surveillance of hardware and human-centered systems. The project resulted in \$20,000 in funding for my PI.

SCH: AIMS-Pro: AI-Driven Multimodal Stress Profiling of Expert and Novice Practitioners in Middle School Math and Nursing

- Supported proposal development for an interdisciplinary project focused on AI-driven multimodal stress profiling across educational and nursing contexts. This proposal did not result in any funding.

A Secure and Scalable Electronics Supply Chain Framework for Authenticating and Tracking ICs Using Hierarchical Bloom Filter

- Contributed to the development of a white paper outlining a secure and scalable electronics supply chain framework for authenticating and tracking integrated circuits using hierarchical Bloom filter-based methods. This proposal did not result in any funding.

TECHNICAL SKILLS

Python, MATLAB, SQL, Machine Learning, Deep Learning TensorFlow, PyTorch, Git, AWS, AWS S3, SageMaker, AWS Bedrock, Power BI, Tableau, Arduino, C/C++, Java, Neural Networks, Visio, LLM, Flask, Streamlit, LangChain, R, MySQL, Postgres, JavaScript, TypeScript, Rust, HTML, CSS, Apache PySpark, MongoDB, Deep Learning, Computer Vision, Matplotlib, Plotly, Google Looker Studio, AWS Elastic MapReduce, Iceberg.

AWARDS

- 1st Runner-up, Poster Competition** 2025
FBI 2025 | KU Cybersecurity Conference, University of Kansas, Kansas, USA.
- 1st Runner-up, Pinata Challenge** 2025
KU Blockchain Institute Presents Hack-Midwest Block-A-Thon, University of Kansas, Kansas, USA.
- 2nd Runner-up, General Track – Best Hack** 2025
KU Blockchain Institute Presents Hack-Midwest Block-A-Thon, University of Kansas, Kansas, USA.
- 1st Runner-up, Poster Presentation Graduate Competition** 2025
ISRS Research Symposium, University of Kansas, Kansas, USA.
- 1st Place, Best-in-Show** 2024
Hack-Midwest, Kansas, USA.
- 1st Place, Poster Competition** 2024
FBI 2024 | KU Cybersecurity Conference, University of Kansas, Kansas, USA.
- Summa Cum Laude** 2023
24th Convocation, North South University, Dhaka, Bangladesh.
- 1st Place, Microelectronics Security Challenge (Supply Chain Track)** 2023
Hardware Oriented Security and Trust Conference, HOST, San Jose, California.
- David D. and Mildred H. Robb Award** 2023
Department of EECS, University of Kansas, Lawrence, KS, USA.
- NSF Travel Grant** 2023 and 2022
Hardware Oriented Security and Trust Conference, HOST, Washington, DC and San Jose, USA.
- Best Member, Team Provision** Spring 2021
ACM Student Chapter, North South University.
- 1st Place, Presentation Contest** 2021
ACM Student Chapter, North South University.
- Admission Test Scholarship, Tuition Fee Waiver** Spring 2018
Financial Aid Office, North South University.

14. **Bangladesh Government Scholarship**
Higher Secondary School Certificate Exam, Dhaka Education Board.

2017

ORAL/ POSTER PRESENTATIONS

1. **A Dynamic Hierarchical Bloom Filter-based Framework for Scalable Biometric Authentication Systems**
International Joint Conference on Biometrics, Osaka, Japan, September 2025.
2. **A Hierarchical Bloom Filter-based Framework for Scalable Authentication and Tracking of ICs**
I2S, Raymond Nichols Hall, February 2024.
3. **Secure Efficient and Scalable Framework for Fast Human Surveillance and Verification**
ISRS Research Symposium, Kansas Union, 2025.
4. **A Hierarchical Bloom Filter-based Framework for Scalable Authentication and Tracking of ICs**
ISRS, Kansas Union, February 2025.
5. **Secure Efficient and Scalable Framework for Fast Human Surveillance and Verification**
FBI | KU Cybersecurity Conference, Kansas Union, February 2024.
6. **Secure Efficient and Scalable Framework for Fast Human Surveillance and Verification**
Graduate Engineering Association, University of Kansas, February 2024.
7. **FarmWise: Your Farming Companion**
HackMidwest Hackathon, Kansas City, October 2024.
8. **Autonomous Screed Controller**
The Craig and Diane Martin National Center for Construction Safety Board Advisory Meeting, Kansas City, 2025.
9. **Secure Efficient and Scalable Framework for Biometric Authentication Systems**
FBI | KU Cybersecurity Conference, Kansas Union, April 2024.
10. **Zk-PoH: Zero Knowledge Proof of Humanness**
KU Blockchain Institute Presents Block-A-thon, University of Kansas, March 2025.
11. **Agentic Appointment Creator for Patients**
MLH Hack KU, University of Kansas, May 2025.
12. **Bio-BloomChain: A Probabilistic Smart Contract for Secure and Scalable Decentralized Biometric Authentication Devices**
FBI | KU Cybersecurity Conference, March 2026.
13. **A Dynamic Hierarchical Bloom Filter-based Framework for Scalable Biometric Authentication Systems**
GEA Graduate Showcase, University of Kansas, September 2026.

LEADERSHIP EXPERIENCE

ACM Student Chapter, Bangladesh

Dec. 2020 – Dec. 2021

Role: In-charge, Research & Development Team

- Promoted members' research and development skills by organizing workshops on cutting-edge technologies and managing end-to-end website and social media operations.
- Fostered problem-solving ability and coding proficiency by acting as a programming contest problem setter and arranging hackathons focused on real-world challenges, collaboration, and rapid prototyping.

INVITED REVIEWER

- IEEE Internet of Things Journal, 2025
- Nature Scientific Reports.

GUEST LECTURER

- EECS 330 – Data Structures and Algorithms
- EECS 700 – Computer Vision

COLLABORATIONS

- Prof. Tamzidul Hoque, Ph.D., University of Kansas, Lawrence, KS, USA
- Prof. Alexandra Kondyli, Ph.D., University of Lawrence, Kansas, KS, USA
- Prof. Domenic Forte, Ph.D., University of Florida, Gainesville, FL, USA
- Prof. Prabuddha Chakraborty, Ph.D., University of Maine, Orono, ME, USA
- Dr. Aimon Rahman, Ph.D., Amazon Inc., Seattle, WA, USA

REFERENCES

Dr. Sumaiya Shomaji

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Dr. Tamzidul Hoque

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Dr. Prabuddha Chakraborty

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